

Hypotheses and measurable outcomes

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

A literature comparison matrix

Without opening the old lesson, explain one key idea from A literature comparison matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Methods results and limitations

Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A question that evidence can answer

Without opening the old lesson, explain one key idea from A question that evidence can answer. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Hypotheses and measurable outcomes. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Hypotheses and measurable outcomes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Before running 12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

After seeing faster responses, the team changes its primary claim from support improvement to speed improvement. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A hypothesis can fail honestly; changing it after results does not erase the original test. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Before running 12 cases, you expect B support to exceed A. Write primary metric and one falsifying result.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled hypotheses and measurable outcomes evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of hypotheses and measurable outcomes. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Hypotheses and measurable outcomes

Write a testable hypothesis before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

Ten cases are reserved. Unsupported means a factual claim absent from the approved card. Prediction: B has fewer unsupported cases than A.

Hypotheses and measurable outcomes

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name the directional claim

The prediction concerns unsupported cases, not writing elegance.



2. Define scoring first

Each answer either contains an unsupported claim or does not under the fixed rubric.



3. Freeze the comparison

Run both versions on the same cases after recording the hypothesis.

CHANGE ONE THING

B sounds friendlier but has the same unsupported count as A.

Explain the changed result using the matching lesson.

Name the predicted outcome and how unsupported is scored. If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a testable hypothesis before scoring. A team predicts that a retrieval rule will reduce unsupported answers on a fixed validation set.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Run both versions on the same cases after recording the hypothesis.

1. Name the directional claim
2. Freeze the comparison
3. Define scoring first

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Freeze the comparison Run both versions on the same cases after recording the hypothesis. A hypothesis makes an expected pattern explicit before seeing outcomes. The measured definition must be clear enough for another reviewer to apply.

Practice 2: explain the deciding evidence

Changed-case practice: B sounds friendlier but has the same unsupported count as A. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. The stated improvement hypothesis is not supported on these cases.
3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The stated improvement hypothesis is not supported on these cases. Friendliness is a different outcome from the predefined unsupported-case count.

Repair a missing building block

Review A literature comparison matrix (ai-7/6). Goal: Explain the mechanism and terminology of a literature comparison matrix. Short practice: Explain why 8/10 indoors and 16/20 outdoors share a rate but not a controlled comparison. Which condition column must the literature matrix preserve?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the mechanism and terminology of methods results and limitations. Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor evidence in their proper categories. Why cannot the result establish that B is best for outdoor farming?. Return to this week and explain what you changed.

Review A question that evidence can answer (ai-7/1). Goal: Explain the mechanism and terminology of a question that evidence can answer. Short practice: Which proposed diagram question has a measurable outcome? Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

